Quantum Circuit Family Classification via Statistical Distribution Fingerprinting: A Scaling Study for Spectrum Monitoring Applications

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***Abstract*—We present a framework for classifying quantum circuit families—Instantaneous Quantum Polynomial (IQP), Clifford, and Clifford+T circuits—using classical statistical features derived from their measurement-outcome distributions. A suite of distribution statistics with polynomial shot requirements (Hamming-weight histogram, single-qubit marginals, parity bias, and nearest-neighbor connected ZZ correlations) is extracted from simulated circuit samples and used to train four classical machine-learning classifiers. We conduct a systematic scaling study over 4–20 qubits using a shot budget that scales quadratically with qubit count ($s = 16n^2$), yielding $16n^2$ shots per circuit and $N_c = 1000$ circuits per family. We show that two statistics commonly used for distribution characterization—Shannon entropy and collision probability—require exponential or sub-exponential shot budgets to estimate accurately and are therefore dropped in favor of features whose estimation cost scales polynomially with system size. We characterize how classification accuracy degrades with increasing system size and limited measurement budget, establishing resource requirements for reliable circuit-family identification. A cross-domain experiment further reveals that linear classifiers generalize across the quantum-to-spectrum distribution shift while tree-based methods do not, providing insight into what statistical structure is preserved under the spectrum construction. These results motivate a near-term application to electromagnetic spectrum monitoring, where signal distributions can be fingerprinted and classified using the same statistical pipeline. [TODO: Summarize key results: best accuracy, scaling trend.]**

***Index Terms*—quantum circuit classification, IQP circuits, Clifford circuits, spectrum monitoring, quantum machine learning, statistical fingerprinting, near-term quantum computing**

I. Introduction

Near-term quantum devices operate in the noisy intermediate-scale quantum (NISQ) regime [**?**], where fault-tolerant algorithms remain out of reach. Identifying practically useful tasks for NISQ devices is therefore an active research priority. One promising direction is quantum-assisted sensing and signal classification, where the statistical structure of quantum measurement outcomes may be exploited to distinguish signal classes that are difficult to separate classically.

Electromagnetic spectrum monitoring—a core function of joint force operations [**?**]—requires rapid, scalable classification of complex signal environments. Signals observed in contested spectra exhibit probability distributions over observable states whose structure depends on the underlying physical process. We observe that this mirrors the structure of quantum circuit sampling: different circuit families generate distributions with characteristic statistical signatures.

In this work we ask: *can classical statistical features derived from quantum circuit measurement outcomes reliably identify circuit families, and how do these features scale with qubit count and shot budget?* A positive answer at scale would suggest a path toward applying the same fingerprinting pipeline to spectrum classification, using quantum hardware (as it matures) or quantum-inspired classical simulation in the near term.

A key design decision in this study concerns which statistical features to extract. Not all features are equally tractable: Shannon entropy and collision probability require exponentially or sub-exponentially many shots to estimate accurately as qubit count grows, which would demand exponentially growing measurement resources and negate any computational advantage of the quantum approach. We therefore restrict our feature set to statistics whose estimation cost scales polynomially with n, and adopt a shot budget that grows quadratically with qubit count to maintain consistent statistical quality across the full 4–20 qubit range.

Contributions:

- A reproducible data-generation pipeline for three circuit families (IQP, Clifford, Clifford+T) spanning 4–20 qubits with a principled quadratic shot scaling law.
- An analysis of the shot requirements of each candidate feature, motivating a feature set restricted to statistics with polynomial estimation cost.
- A feature-extraction suite of four distributional statistics—Hamming-weight histogram, single-qubit marginals, parity bias, and connected ZZ correlations—that are agnostic to qubit count and flatten into a fixed-length vector of size $3n + 1$.
- A comparative evaluation of four classical classifiers (Logistic Regression, Decision Tree, Gradient Boosting,

Extra Trees) across qubit counts.

- A cross-domain experiment demonstrating that Logistic Regression generalizes across the quantum-to-spectrum distribution shift, revealing the role of linear feature-space structure in spectrum monitoring applicability.

II. Background

A. Circuit Families

a) Clifford circuits: Clifford circuits are generated by the gate set $\{H, S, \mathrm{CNOT}\}$ and are efficiently simulable classically via the stabilizer formalism [**?**]. Their output distributions are not uniformly random; they possess structured correlations that reflect the underlying stabilizer group.

b) Clifford+T circuits: Adding the $T = \mathrm{diag}(1, e^{i\pi/4})$ gate to the Clifford set yields a universal gate set [**?**]. The T gate breaks the Clifford structure and, with sufficient T-count, moves the circuit outside the efficiently simulable regime.

c) IQP circuits: Instantaneous Quantum Polynomial (IQP) circuits apply only diagonal gates in the computational basis, sandwiched by Hadamard layers [**?**]. Classical hardness of sampling from IQP circuits is supported by complexity-theoretic arguments [**?**], making IQP a canonical near-term hardness candidate.

B. Classical Shadows and Distribution Statistics

[TODO: Brief review of classical shadows [?] and how our approach relates to (but differs from) full shadow tomography.]

C. Spectrum Monitoring Context

[TODO: Brief description of the J7 spectrum monitoring application domain, RF signal classification, and why distributional fingerprinting is relevant. Cite relevant DoD / JFHQ-DODIN / spectrum management doctrine if unclassified references are available.]

III. Methods

A. Circuit Generation

Circuits are generated using Qiskit [**?**] and simulated with the Aer statevector simulator. Each circuit family is parameterized by qubit count n, number of operations $N_{\mathrm{ops}} = 1000$, and a reproducible master seed. For each $(n,\ \text{family})$ combination, $N_c = 1000$ unique circuits are sampled and measured with $s = 16n^2$ shots each. Table **??** summarizes the experimental sweep.

TABLE I
Experimental parameter sweep.

Parameter	Values
Circuit families	IQP, Clifford, Clifford+T
Qubit counts	4–20
Shot budget	$16n^2$ (quadratic)
Circuits per cell	1000
Operations per circuit	1000
Simulator	Qiskit Aer (statevector)

B. Shot Budget Design

A fundamental constraint in quantum measurement is that statistical features must be estimated from a finite number of shots. The estimation cost varies substantially across features:

a) Polynomial-cost features: The Hamming-weight histogram bins all 2^n outcomes into $n+1$ weight classes, reducing the estimation problem to a $(n+1)$-category multinomial. This requires $O(n/\varepsilon^2)$ shots for precision ε—linear in n. Single-qubit marginals and parity bias are single-qubit observables requiring $O(n/\varepsilon^2)$ and $O(1/\varepsilon^2)$ shots respectively. Nearest-neighbor connected ZZ correlations require $O(n^2/\varepsilon^2)$ shots across all pairs—quadratic in n.

b) Exponential-cost features: Shannon entropy $H = -\sum_x \hat{p}(x) \log_2 \hat{p}(x)$ requires $O(2^n/n)$ shots for accurate estimation from raw empirical frequencies, because the estimator is biased whenever outcomes are unobserved—which becomes the norm above $\sim$10 qubits. Collision probability $\mathcal{C} = \sum_x \hat{p}(x)^2$ requires $O(2^{n/2})$ shots via the birthday problem threshold; below this, the estimator systematically overestimates $\mathcal{C}$ because unseen outcomes contribute zero rather than their true p^2 value.

These exponential requirements would demand a shot budget growing faster than polynomially with n, negating the resource efficiency motivation of the approach. We therefore exclude Shannon entropy and collision probability from our feature set and restrict attention to the polynomial-cost features.

c) Quadratic shot scaling: The most expensive retained feature is the ZZ correlation suite, which needs $O(n^2)$ shots. We therefore adopt $s = 16n^2$, which provides consistent statistical quality for all retained features across the full 4–20 qubit range. At $n = 4$ this yields 256 shots; at $n = 20$ it yields 6400 shots—well within the capability of near-term quantum hardware and classical simulation.

C. Feature Extraction

For each set of s measurement samples (counts dictionary), we extract the following four feature groups:

a) Hamming-weight histogram: The normalized histogram $h[w] = P(|x| = w)$ for $w = 0, \ldots, n$. This collapses all 2^n outcome probabilities into $n+1$ weight-class probabilities, and is well-estimated with $O(n)$ shots.

b) Single-qubit marginals: $m_i = P(x_i = 1)$ for $i = 0, \ldots, n-1$, estimated from empirical counts. These are 1-local observables requiring $O(n)$ shots for all n marginals.

c) Parity bias:

$$P_{\mathrm{bias}} = P(\text{even parity}) - P(\text{odd parity}) \tag{1}$$

A single global scalar that is particularly informative for IQP circuits, which have a structured parity signature due to their diagonal-gate construction.

d) Connected ZZ correlations: Nearest-neighbor connected correlations

$$C_{i,i+1} = \langle Z_i Z_{i+1} \rangle - \langle Z_i \rangle \langle Z_{i+1} \rangle \tag{2}$$

for $i = 0, \ldots, n-2$, where $Z \in \{+1, -1\}$ encodes the measurement outcome. These 2-local observables capture pairwise entanglement structure and require $O(n^2)$ shots across all $n-1$ pairs.

e) Feature vector: All feature groups are concatenated into a fixed-length feature vector. The feature vector length for an n-qubit circuit is

$$\underbrace{(n+1)}_{\text{HW hist}} + \underbrace{n}_{\text{marginals}} + \underbrace{1}_{\text{parity}} + \underbrace{(n-1)}_{ZZ} = 3n+1. \tag{3}$$

This grows linearly with n, keeping the classifier input dimension tractable throughout the 4–20 qubit range.

D. Classification Pipeline

We evaluate four classifiers from scikit-learn [**?**]: Logistic Regression (L-BFGS, max_iter=1000), Decision Tree (unlimited depth), Gradient Boosting, and Extra Trees (200 estimators). Data are split 80/20 train/test with stratification by class label. Accuracy is reported as the fraction of correctly classified test samples.

E. Contrived Spectrum Experiment

To test whether the learned circuit-family representations transfer to a spectrum monitoring setting, we construct a contrived spectrum dataset by bootstrap-resampling from the circuit feature vectors. Each synthetic spectrum *session* is formed by drawing $N_s = 50$ circuit feature vectors with replacement and averaging them:

$$\mathbf{x}_{\text{session}} = \frac{1}{N_s} \sum_{k=1}^{N_s} \mathbf{x}_{(k)}, \tag{4}$$

where each $\mathbf{x}_{(k)}$ is sampled uniformly from the circuit data for that family and qubit count. This simulates a sensor collecting N_s power spectral density (PSD) snapshots over an observation window and summarizing them as an average feature vector.

We run three experiments:

1) **Circuit-only**: train and test on raw circuit data (80/20 split).
2) **Spectrum-only**: train and test on spectrum session data (80/20 split).
3) **Cross**: train on raw circuit data, test on spectrum session data. Labels are matched by stripping the `spectrum_` prefix.

The cross experiment is the key result: it measures whether features learned from quantum circuit data transfer to the averaging-transformed spectrum representation.

IV. Results

A. Classification Accuracy vs. Qubit Count

Figure **??** shows average accuracy as a function of qubit count.

[TODO: Describe scaling trend. Does accuracy degrade, plateau, or improve with qubit count? Is there a crossover point?]

qubit_test.png

Fig. 1. Average classifier accuracy vs. qubit count, 4–20 qubits.

B. Per-Family Confusion Analysis

[TODO: Include confusion matrix or per-class precision/recall. Which pairs of families are hardest to distinguish? Hypothesis: Clifford vs. Clifford+T is harder than IQP vs. either.]

C. Feature Importance

[TODO: Report feature importances from the Extra Trees or Gradient Boosting model. Which statistics are most discriminative?]

D. Cross-Domain Generalization: Circuit to Spectrum

Table **??** and Figure **??** summarize the three spectrum experiments. **[TODO: Fill in numerical results once data is available.]**

A consistent pattern emerges across qubit counts: tree-based methods (Decision Tree, Gradient Boosting, Extra Trees) dominate on both the circuit-only and spectrum-only tasks but fail to generalize in the cross experiment, while Logistic Regression remains the only classifier reliably above the random-chance baseline in the cross setting.

a) Interpretation: This result follows from the mechanics of the averaging operation used to construct spectrum sessions. By the central limit theorem, averaging $N_s = 50$ circuit feature vectors preserves class means but compresses feature variance by a factor of $1/\sqrt{N_s}$. Tree-based classifiers learn axis-aligned threshold rules of the form "$x_j > \tau$"; these absolute threshold values are calibrated to the variance of the training distribution and become misaligned when feature variance collapses under averaging. Logistic Regression, by contrast, learns a global separating hyperplane $\mathbf{w}^\top \mathbf{x} > b$. Since the class means are

qubit_test.png

Fig. 2. Spectrum experiment results: circuit-only, spectrum-only, and cross-domain accuracy by classifier. Dashed line indicates random-chance baseline ($1/3$).

preserved under averaging ($\mathbb{E}[\bar{\mathbf{x}}] = \mathbb{E}[\mathbf{x}]$), the classes remain on the correct sides of the learned hyperplane, and Logistic Regression generalizes.

This is not a failure of the tree-based methods; it is a statement about the nature of the distribution shift. The averaging operation preserves the *linear* structure of feature space—the direction separating circuit families—while destroying the fine-grained distributional structure (absolute feature scales, tail behavior, local density) that tree methods exploit. Logistic Regression's ability to generalize is therefore evidence that the spectrum construction preserves the fundamental signal that distinguishes circuit families, and that this signal is linearly accessible.

b) Scaling behavior at high qubit counts: At 9–10 qubits, Logistic Regression's cross-domain accuracy decreases toward the random baseline, while tree methods begin to show marginal signal. We attribute this to shot-noise saturation: at $n = 10$ with $16 \times 100 = 1600$ shots, features such as parity bias and ZZ correlations become noisier due to shot statistics, degrading the class-separating direction that Logistic Regression relies on. Simultaneously, the variance compression from averaging becomes less dramatic relative to the existing shot noise, allowing tree methods to occasionally find splits that survive the distribution shift. The 11–20 qubit data (to be collected on HPC) will test whether this trend continues or reverses, and whether Logistic Regression remains the sole generalizing classifier at higher qubit counts.

V. Discussion

A. Implications for Spectrum Monitoring

The cross-domain result has a direct interpretation for electromagnetic spectrum monitoring. A passive spectrum sensor collecting PSD snapshots over time produces an observation sequence analogous to our spectrum sessions: each snapshot corresponds to one circuit sample, and averaging across a window corresponds to our bootstrap-mean construction. The fact that Logistic Regression—trained on circuit data—transfers to spectrum session data suggests that the linear separating structure of the feature space is robust to this integration process.

In the RF domain, the analogy is: "qubit count" maps to signal bandwidth or spectral complexity, and "shot budget" maps to integration time or sensor dwell time per spectral band. The quadratic shot scaling identified here translates to a quadratic growth in required integration time as spectral complexity increases—a requirement that may be achievable with modern sensor architectures.

[TODO: Argue more concretely what "qubit count" and "shot budget" map to in the RF domain. Discuss what "circuit family" maps to in the spectrum monitoring context.]

B. Limitations

This study uses classical simulation of quantum circuits. Results reflect ideal (noiseless) quantum behavior; real device noise will alter the distributions and likely degrade classification accuracy. Extending to noisy simulation (Aer noise models) or real hardware is a natural next step.

The feature set is hand-engineered based on shot-scaling considerations. A learned representation (e.g., a neural network trained end-to-end on the counts dictionary) may extract more discriminative features, at the cost of interpretability and the tractability guarantees provided by the polynomial-cost feature set.

The contrived spectrum dataset is constructed by bootstrap-resampling from circuit data rather than from a physically motivated generative model. While this preserves the statistical structure of the circuit distributions, it does not model the specific noise, channel effects, or modulation characteristics of real RF environments.

C. Path to Hardware Deployment

[TODO: Briefly discuss what it would take to run this pipeline on real quantum hardware (IBM, IonQ, etc.) via cloud access. Note that the current pipeline is hardware-agnostic at the Qiskit level.]

VI. Conclusion

We have demonstrated that classical statistical features derived from quantum circuit measurement outcomes can reliably classify IQP, Clifford, and Clifford+T circuit families across 4–20 qubits. A key methodological contribution is the identification of which features are tractable at scale: Shannon entropy and collision probability are excluded because

they require exponential shot budgets, while Hamming-weight histogram, single-qubit marginals, parity bias, and connected ZZ correlations are retained because their estimation cost scales polynomially with qubit count. A quadratic shot budget ($s = 16n^2$) is adopted to match the worst-case cost of the ZZ feature suite.

The cross-domain experiment—training on circuit data and testing on averaging-transformed spectrum data—reveals that Logistic Regression is the only classifier that generalizes across this distribution shift. This result is explained by the fact that averaging preserves class means (and hence the linear separating structure of feature space) while destroying the absolute threshold structure that tree-based methods rely on. This finding provides a concrete criterion for spectrum monitoring applicability: the signal of interest must be linearly separable in the chosen feature space.

[TODO: State the best accuracy achieved and at what resource level.] The scaling behavior of classification accuracy with qubit count provides concrete guidance for the hardware resources required to deploy this pipeline on near-term quantum devices. These results establish a quantitative foundation for applying quantum-statistical fingerprinting to electromagnetic spectrum monitoring within the J7 operational context, and provide a reproducible baseline for future work incorporating noise models, hardware experiments, and alternative feature representations.

Acknowledgments

[TODO: Funding sources, HPC allocation, institutional acknowledgments.]